Aayush Kumar Bhat

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Portfolio: aayush-portfolio-390a.onrender.com

Summary

Software engineer with production experience in telemetry-driven backend systems, streaming pipelines, and deployment automation (Python, Java, C++, SQL); ships data platforms on Azure and AWS and diagnoses root causes from logs, metrics, and data-quality signals.

Experience

Nagarro Gurugram, India
Associate Software Engineer Sep 2026 – Present
Associate Software Engineer Trainee Jan 2026 – Sep 2026

- Processed daily streaming telemetry events with **PySpark** and built window-function validation across **150+** streams with watermark-based correctness.
- Managed production deployment pipelines with **GitHub Actions** CI/CD; used **Azure Monitor** for system health and log-based diagnostics on automated batch jobs across **85+** tables.
- Co-built an **Azure Data Factory** and **Azure SQL** medallion POC for simulated EV-charger telemetry, with **8** pipelines, a **12**-rule exception layer and packet-gap detection, **15** data-quality checks, and a **6**-page **Power BI** report.
- Co-built a **PySpark** and **Delta Lake** medallion POC for simulated HVAC telemetry (**81** chillers, **140K** readings/day), adding rule-based data quality with quarantine and watermark-driven incremental runs that populate **24** Gold tables.
- Built backend services and REST APIs using **Python** and **FastAPI** with **DuckDB** zero-copy OLAP queries over **SQL** data.

Skills

Languages & Databases: C++, Python, Java, SQL, TypeScript, JavaScript, PostgreSQL, MongoDB, Redis, SQLite, non-relational databases, T-SQL
Web Technologies: Next.js, FastAPI, Node.js, Express.js, React, Socket.IO, REST APIs, GraphQL, HTML, CSS, Tailwind CSS, Three.js, Prisma, NextAuth
Cloud & Platforms: Azure Data Factory, Databricks, Apache Kafka, Apache Spark (PySpark), Delta Lake, Unity Catalog, DuckDB, AWS, AWS Lambda, Amazon EMR, OCI, Hugging Face, Render, Neon, Azure SQL, Power BI
Core Concepts: Data Structures & Algorithms (DSA), Data Modeling, Data Pipelines, Streaming Systems, Consumer Groups, Asynchronous Programming, Event-Driven Architecture, Columnar Data Models, Microservices, ETL, Hadoop, Hive, NLP, MinHash, TF-IDF, Parquet, PyArrow, OAuth 2.0, OpenID Connect, JWT, RBAC, Argon2id, TOTP, CSRF, Distributed Systems, Data Warehousing, Data Governance, Medallion Architecture, Data Quality
Testing, Tools & Observability: Docker, GitHub Actions, CI/CD, Automated Testing, Azure Monitor, Application Insights, Log Analysis, Git, pytest

Projects

Globe News Atlas | *Live Demo*
Python, FastAPI, React, TypeScript, SQL, Parquet, PyArrow, SQLite, Three.js, GitHub Actions, Docker, pytest

- Built a full-stack world-news atlas ingesting ten public sources into a bronze/silver/gold **Parquet** data lake on **Hugging Face**, serving **400-event** 24-hour snapshots through a **FastAPI** service and a React/Three.js WebGL globe.

TaskFlow – Chat-Native Work OS | *Live Demo* | *GitHub*
Next.js 14, React, TypeScript, Tailwind CSS, PostgreSQL, Prisma, Socket.IO, NextAuth, Docker, Render, Neon

- Built and deployed a chat-native work OS spanning tasks, notes, mind maps, calendar, focus sessions, and chat across **62 Prisma models**, as a single **Docker** service on Render with Neon PostgreSQL and startup migrations.
- Engineered real-time messaging with **Socket.IO** rooms over asynchronous event handlers, including threads with live reply counts, reactions, typing indicators, presence, pinned decisions, and read state, backed by REST APIs and Prisma.

HVAC Telemetry Lakehouse | *Source Code*
Apache Kafka, confluent-kafka, PySpark, Delta Lake, DuckDB, FastAPI, Server-Sent Events, GitHub Actions

- Built a public proof of concept streaming simulated HVAC telemetry through **Kafka** over TLS into a **Delta Lake** medallion pipeline in **PySpark**, serving the gold layer to a **FastAPI** read path that runs **DuckDB** queries over Arrow, with **Z-Ordering** on the tables the API filters.
- Ran two **Kafka** consumer groups with different offset policies, a live dashboard on `latest` and a batch drain on `earliest`, and instrumented every stage so each API response carries its own `latency_ms`.

Education

University School of Information, Communication and Technology Delhi, India
*Master of Computer Applications, Software Engineering; GPA: **8.99/10*** Aug 2024 – May 2026

Integrated Institute of Technology Delhi, India
*Bachelor of Computer Applications; GPA: **9.01/10*** Aug 2020 – May 2023

Achievements

- LeetCode contest rating of **2,103** (Top **1.56%** globally), **Knight** status, and **752** problems solved.
- Secured All India Rank **1,030** in the NIMCET 2024 national entrance examination among **50,000+** candidates.

Certifications

- **Oracle Cloud Infrastructure 2025 Foundations Associate** | *Credential* (Oct 19, 2025; valid to Oct 19, 2027).
- **Complete React Developer (w/ Redux, Hooks, GraphQL)** | *Credential* (Jul 10, 2025; 42 hours).
- **HackerRank REST API (Intermediate)** | *Credential* (Oct 7, 2025).